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Compositional Phase Separation in $La_{2-x}Ba_xCuO_y$ near the Optimum Composition for Superconductivity

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Compositional phase separation was observed in $La_{2-x}Ba_xCuO_y$ near the optimum composition ($x\sim0.16$) for superconductivity with a decisive miscibility gap by means of X-ray diffraction measurements. The phase-separated samples do not show intrinsic superconductivity. Therefore, the superconductive compound in this system is not in a chemical equilibrium state at low temperatures, but is a high-temperature phase obtained by quenching to room temperature. The critical temperature of the phase separation T_p takes a maximum value of 950°C at $x\sim0.16$, suggesting that phase instability is crucial for high-T_c superconductors. The end composition of the miscibility gap in the Ba-poor composition region was found to be 0.12, which is a singular point from crystallographic and band structural standpoints.

[phase separation, high-T_c superconductivity, $La_{2-x}Ba_xCuO_4$]

Since the discovery of superconductivity with the fairly high transition temperature T_c in the cuprate $La_{2-x}Ba_xCuO_4$ system,[1] many investigations have been carried out on the so-called 214 system with the K_2NiF_4- or Nd_2CuO_4-type structure in order to elucidate the mechanism of the high-T_c superconductivity. To date, several characteristic features have been found, as follows.

(1) The transition temperature T_c for all the doped 214 systems shows a characteristic composition dependence in an essentially common manner: T_c takes a maximum around $x\sim0.15$, decreases with both increasing and decreasing x and disappears for $x<0.06$ and $x\geqq0.25$.[2,3]

(2) Phase instability appears at the composition for the maximum T_c in typical 214 systems, *e.g.*, $La_2CuO_{4+\delta}$, $Nd_{2-x}Ce_xCuO_4$ and $La_{2-x}Sr_xCuO_4$, as found and/or pointed out by Jorgensen *et al.*[4,5] and Lightfoot *et al.*[6]

(3) Successive structural phase transitions occur in La-214 systems around $x\sim0.12$ (distinctly in the $La_{2-x}Ba_xCuO_4$ system), accompanied with a large dip in T_c. This composition has been recognized as a singular point from standpoints of both crystallography and electronic state theory.[7–9]

These features should be further investigated because they must be related to the mechanism of high-T_c superconductivity.

We carried out systematic investigations of the Meissner effect using powdered samples of $La_{2-x}Sr_xCuO_y$ and $La_{2-x}Ba_xCuO_y$ systems, and concluded that T_c does not change through out the whole composition range, and that the Meissner volume fraction f is dependent on x and takes a maximum at $x\sim0.16$.[10,11] Recently, we carried out systematic investigations of the phase separation of this system using the X-ray diffraction (XRD) technique. As a result, we have succeeded in observing a distinct miscibility gap around the Ba concentration showing maximum T_c. In this paper, we report the newly found compositional phase separation in $La_{2-x}Ba_xCuO_4$ and discuss the relationship of phase separation with the superconductivity and also with the singular composition of $x=0.12$.

The samples were synthesized through solid-state reactions of La_2O_3, $SrCO_3$ and CuO of 99.99% purity. Mixed powders were

ground, pelletized and calcined in air at various temperatures T_s (860~1050°C) for two days followed by quenching. This process was repeated until the XRD pattern no longer changed. We call this usual preparation procedure preparation A. For the preparation of samples with high x at low T_s ($x>0.2$ and $T_s<960$°C), foreign phases such as $La_4Ba_2Cu_2O_y$ and $La_2BaCu_2O_6$ in addition to the K_2NiF_4-type compounds were observed in small amounts by means of XRD. However, this may not be caused by a solubility limit, but may be a matter of reaction rates and/or reaction paths. To avoid this complication, we prepared samples with high x at low T_s using the following procedure (preparation B). First, the starting compounds with the compositions of $x=0.04$ and 0.30 were prepared at 1050°C using preparation A, and then an appropriate amount of two samples was mixed and fired at T_s. Ultimately, we succeeded in synthesizing samples without any foreign phases down to 860°C. The attainment of the chemical equilibrium state was verified by the annealing time dependence of XRD patterns: in the case of $T_s=880$°C, for example, it took a few days for samples formed using preparation A to attain the chemical equilibrium state, and about 60 days for those formed using preparation B. Here, we have about a two-order difference in the time scale, which can be understood by considering the difference between the powder radii in these preparations if we assume the diffusion controlling process: the order of the radius is 0.1 μm in preparation A and 1 μm in preparation B. Fick's second law indicates that $x/\sqrt{t}=$constant (x and t being the length and the time of diffusion). Therefore, a one-order difference in x implies a two-order difference in t (the chemical diffusion constant of La^{3+} and Ba^{2+} being estimated as on the order of 10^{-10}~$^{-13}$ cm^2/s at 880°C). For the following experiments, the powdered samples with grain radii of less than 10 μm were used, with annealing in flowing O_2 gas at temperatures lower than 600°C to release the distortion involved in quenching.

The obtained samples were checked by means of XRD utilizing monochromatized Cu-Kα characteristic X-rays with the step-scanning technique. The superconducting properties were characterized by measuring the ac susceptibility (ac-χ) using the Hartshorn bridge method under a field of 100 mOe oscillating at 270 Hz.

Typical XRD patterns around $2\theta=41$ degrees for several samples in the equilibrium state with different Ba concentrations prepared at $T_s=880$°C are presented in Fig. 1, in which all the diffraction lines originate from the Kα_1, i.e., those from the Kα_2 are subtracted by numerical analysis. One finds that line splittings occur in both the (006) and (105) peaks in the concentration range of $0.12<x\leqq0.26$ which suggests compositional phase separation. It is important to note that the other peaks in the XRD patterns can also be indexed as two phase mixtures with the K_2NiF_4-type structure. The intensity change with Ba concentration of the diffracted peaks for each phase roughly obeys *the lever rule.*

To obtain clear evidence for the existence of compositional phase separation, we carried out a systematic investigation by changing T_s. The typical results for the nominal Ba concentration $x_n=0.18$ are presented in Fig. 2. Line splitting due to the phase separation is observed below $T_s=920$°C. Each diffracted angle for split lines depends on T_s in this figure, while it hardly depends on the nominal Ba concentration x_n at fixed T_s, as shown in Fig. 1. The compositions of separated phases were estimated on the assumption that the lattice parameter vs composition curves at room temperature for the sample prepared above the critical temperature of phase separation T_p, i.e., in the solid solution region, are roughly equal to those for the samples prepared below T_p. The summary of the analysis is presented in Fig. 3. As a result, the distinct miscibility gap can be defined as the shadowed area in Fig. 3. The maximum value of T_p is about 950°C (at $x\sim0.16$).

The superconducting properties were compared among different T_s. In Fig. 4, as typical examples, the values of the volume fraction of the Meissner effect, f, of the samples prepared at 1050°C (□) and 880°C (●) are plotted against x_n. Although the concentration dependence of T_c was also checked, we could not recognize strong dependence of T_c on T_s, as in the case of x-dependence of T_c.[10,11] However,

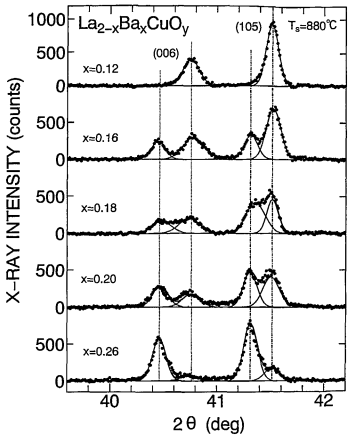


Fig. 1

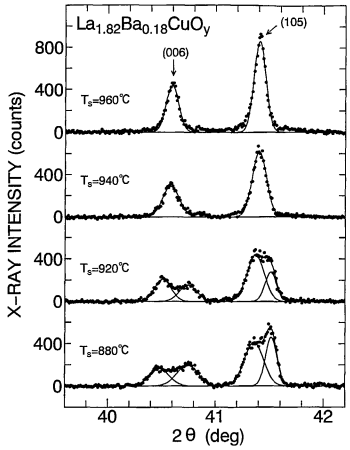


Fig. 2

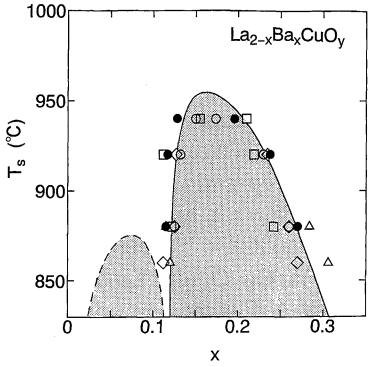


Fig. 3.

Fig. 1. Typical Cu-Kα_1 XRD patterns around $2\theta=41$ degrees for various x synthesized at $T_s=880°C$. The solid curves represent the decomposed diffraction lines analyzed using least-squares fitting.

Fig. 2. XRD patterns for $x=0.18$ synthesized at various T_s.

Fig. 3. Phase diagram presented as a relationship between T_s and two real compositions x into which the nominal composition x_n is separated (see text). ●: $x_n=0.16$, ○: $x_n=0.18$, □: $x_n=0.20$, ◇: $x_n=0.22$, △: $x_n=0.26$.

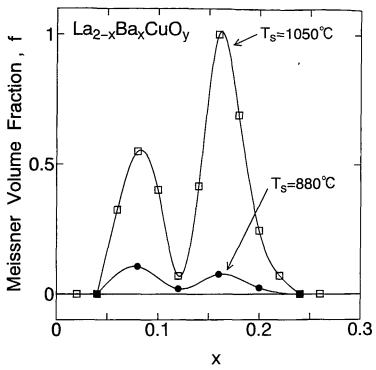


Fig. 4. Meissner volume fraction f plotted against the nominal composition x for the samples synthesized at $T_s=880°C$ (●) and 1050°C (□).

the values of f showed a considerable dependence on both x and T_s, as seen in Fig. 4. The drastic collapse of superconductivity can be observed around $x\sim0.12$. Since the samples with composition $x\sim0.12$ shows structural phase transition at low temperatures (~60 K), it has been recognized that the superconducting state becomes unstable in the low-temperature tetragonal phase.[7-9,11] From this, one concludes that this composition should be a singular point from both crystallographic and band structural standpoints.

Drastic change can be seen in the case of low-T_s synthesis. First, one finds a substantial decrease of f for $0.12<x<0.24$, accompanied with the occurrence of compositional phase separation. This result can be expected because the sample exhibiting compositional phase separation contains two components, neither of which shows superconductivity. One component has the singular composition and the other the composition with the overdoped normal metallic character. This result might have important implications in the mechanism of the occurrence of superconductivity. Related to these results are the theories proposed by Varma[12] and Markiewicz,[13] in which the phase instability in the high-T_c system is discussed on the basis of the charge density wave state or the Van Hove singularity in the band structure, which are characteristic of the two-dimensional nature, as in cuprate systems. They predict compositional phase separation, or phase separation of carriers, the latter of which occurs if diffusion kinetics itself does not allow macroscopic phase separation. The latter phase separation of carriers has already been seen in the $La_{2-x}Sr_xCuO_y$ system with oxygen deficiencies.[14,15] In that case, it was revealed that the microscopic separation between different electronic states was realized, one of which is metallic and supercon-

ducting and the other, insulating and antiferromagnetic.[14,15] The present result is the same as that for former case of macroscopic phase separation, in good correspondence with the other reports of Jorgensen *et al.*[4,5] and Lightfoot *et al.*[6]

In the composition range lower than $x=0.12$, a substantial reduction of f was also observed. This may suggest the possibility that compositional phase separation occurs in this composition range in a slightly lower temperature region than that for $x>0.12$, as speculatively demonstrated in Fig. 3 by a dashed curve, although we could not observe any experimental XRD evidence in this region. The present result may indicate that the composition $x\sim0.12$ corresponds to the line-phase compound which may have some ordered structure, implying that this composition is a singular point, in agreement with the previous results.[7-9,11]

Finally, we must discuss the reversibility of the compositional phase separation, i.e., the phase-separated samples prepared below T_p transform into a single phase upon annealing above T_p, and vice versa. Until the present study, we observed only the former transformation, using XRD. In order to confirm the latter transformation, single-phase samples with $x=0.16\sim0.20$ synthesized at 1050°C were annealed at $800\sim900$°C for $2\sim3$ months. We could not observe the phase separation of these samples using XRD. This result may imply that diffusion kinetics do not allow a phase separation of macroscopic scale detectable using XRD in the present system. The Meissner volume fraction f should be much more sensitive to phase separation than XRD because the coherence length of the superconductivity is on the order of 10 Å in the *a-b* plane in the cuprate system. Recently, we succeeded in observing the latter transformation through measurements of the Meissner volume fraction f with $x=0.16$. The value of f was reduced by 60% upon annealing at 880°C for 30 days. The details of the results will be reported elsewhere after performing systematic investigations.

In conclusion, the present result is the first observation of compositional phase separation having a distinct miscibility gap in the substituted La-214 system, and should support the above-mentioned theories[12,13] of phase instability in high-T_c cuprate systems. The phase-separated sample does not show superconductivity, and the composition (~0.16) which gives the maximum critical temperature of phase separation ($T_p\sim950$°C) corresponds to the optimum one for superconductivity, implying that the phase instability may be necessary for obtaining high-T_c superconductors and be related to the mechanism of the high-T_c superconductivity in cuprate systems.

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